

The **Carvalho (2025)** approximation estimates wavelength **L** of linear surface gravity waves directly from wave period **T** and water depth **d**, without iteration.

The underlying wave dispersion relation is

$$\omega^2 = gk \tanh(kd), \quad \omega = \frac{2\pi}{T}, \quad k = \frac{2\pi}{L}.$$

Define the dimensionless parameter $\alpha = k_0 h$ and the deep-water wavelength L_0 as

$$\alpha = \frac{4\pi^2 d}{gT^2}, \quad L_0 = \frac{gT^2}{2\pi}.$$

The approximate wavelength is

$$L = L_0 \tanh[1.199315(\alpha^{1.047086}) \alpha^{0.499947}]$$

with average error = 0.03% for α in $]0, 2\pi]$ (or $0 < d \leq L_0$) and max error = 0.08% at $\alpha = 1.56$ for any α value.

Each function below has the interface **LWAVE(PERIOD, DEPTH)** and returns L in metres, using the standard acceleration of gravity $g_n = 9.80665 \text{ m/s}^2$.

Use **PERIOD for T** in seconds and **DEPTH for d** in metres. Depth is the positive vertical distance from the still-water surface to the bed.

For $\alpha \geq 20$, the functions return L_0 avoiding overflow when evaluating the nested exponential in deep water.

In Python, the function uses only the standard library:

```
import math
```

```
def LWAVE(PERIOD: float, DEPTH: float) -> float:
    """Return wavelength [m] from period [s] and depth [m]."""
    if (not math.isfinite(PERIOD) or not math.isfinite(DEPTH)
        or PERIOD <= 0.0 or DEPTH <= 0.0):
        raise ValueError("PERIOD and DEPTH must be finite and > 0.")
```

```

g = 9.80665
L0 = g * PERIOD**2 / (2.0 * math.pi)
alpha = 4.0 * math.pi**2 * DEPTH / (g * PERIOD**2)

if alpha >= 20.0:
    return L0

x = 1.199315**(alpha**1.047086) * alpha**0.499947
return L0 * math.tanh(x)

```

In Java 8 or later, save this class as WaveDispersion.java and call WaveDispersion.LWAVE(PERIOD, DEPTH):

```

public final class WaveDispersion {
    public static double LWAVE(double PERIOD, double DEPTH) {
        if (!Double.isFinite(PERIOD) || !Double.isFinite(DEPTH)
            || PERIOD <= 0.0 || DEPTH <= 0.0) {
            throw new IllegalArgumentException(
                "PERIOD and DEPTH must be finite and > 0.");
        }

        final double g = 9.80665;
        final double pi = Math.PI;
        final double L0 = g * PERIOD * PERIOD / (2.0 * pi);
        final double alpha = 4.0 * pi * pi * DEPTH
            / (g * PERIOD * PERIOD);

        if (alpha >= 20.0) {
            return L0;
        }

        final double x = Math.pow(1.199315,
            Math.pow(alpha, 1.047086))
            * Math.pow(alpha, 0.499947);
        return L0 * Math.tanh(x);
    }
}

```

The equivalent C++ function requires C++11 or later:

```

#include <cmath>
#include <stdexcept>

```

```

double LWAVE(double PERIOD, double DEPTH)
{
  if (!std::isfinite(PERIOD) || !std::isfinite(DEPTH)
      || PERIOD <= 0.0 || DEPTH <= 0.0) {
    throw std::invalid_argument(
      "PERIOD and DEPTH must be finite and > 0.");
  }

  const double g = 9.80665;
  const double pi = std::acos(-1.0);
  const double L0 = g * PERIOD * PERIOD / (2.0 * pi);
  const double alpha = 4.0 * pi * pi * DEPTH
    / (g * PERIOD * PERIOD);

  if (alpha >= 20.0) {
    return L0;
  }

  const double x = std::pow(1.199315, std::pow(alpha, 1.047086))
    * std::pow(alpha, 0.499947);
  return L0 * std::tanh(x);
}

```

In Fortran 2008 or later, save the module as wave_dispersion.f90. Import it with use wave_dispersion, only: LWAVE and pass real(real64) arguments. For example, after importing real64 from iso_fortran_env, use LWAVE(9.0_real64, 5.0_real64):

```

module wave_dispersion
  use, intrinsic :: iso_fortran_env, only: real64
  use, intrinsic :: ieee_arithmetic, only: ieee_is_finite
  implicit none
contains
  function LWAVE(PERIOD, DEPTH) result(L)
    real(real64), intent(in) :: PERIOD, DEPTH
    real(real64), parameter :: g = 9.80665_real64
    real(real64), parameter :: pi = acos(-1.0_real64)
    real(real64) :: L, L0, alpha, x

    if (.not. ieee_is_finite(PERIOD) .or. &
        .not. ieee_is_finite(DEPTH) .or. &
        PERIOD <= 0.0_real64 .or. DEPTH <= 0.0_real64) then
      error stop "PERIOD and DEPTH must be finite and > 0."
    end if

```

```

end if

L0 = g * PERIOD**2 / (2.0_real64 * pi)
alpha = 4.0_real64 * pi**2 * DEPTH / (g * PERIOD**2)

if (alpha >= 20.0_real64) then
    L = L0
    return
end if

x = 1.199315_real64**(alpha**1.047086_real64) &
    * alpha**0.499947_real64
L = L0 * tanh(x)
end function LWAVE
end module wave_dispersion

```

In MATLAB, save the following function as LWAVE.m. It accepts scalar numeric inputs and performs the calculation in double precision:

```

function L = LWAVE(PERIOD, DEPTH)
    validateattributes(PERIOD, {'numeric'}, ...
        {'real', 'scalar', 'finite', 'positive'}, ...
        mfilename, 'PERIOD', 1);
    validateattributes(DEPTH, {'numeric'}, ...
        {'real', 'scalar', 'finite', 'positive'}, ...
        mfilename, 'DEPTH', 2);

    PERIOD = double(PERIOD);
    DEPTH = double(DEPTH);
    g = 9.80665;
    L0 = g * PERIOD^2 / (2 * pi);
    alpha = 4 * pi^2 * DEPTH / (g * PERIOD^2);

    if alpha >= 20
        L = L0;
        return
    end

    x = 1.199315^(alpha^1.047086) * alpha^0.499947;
    L = L0 * tanh(x);
end

```

For Excel VBA, paste the following function into a standard module. It returns a numeric result, or #NUM! for non-positive inputs or an arithmetic error. WorksheetFunction.Tanh supplies the hyperbolic tangent:

Option Explicit

```
Public Function LWAVE(ByVal PERIOD As Double, _  
    ByVal DEPTH As Double) As Variant  
    Const g As Double = 9.80665  
    Const PI As Double = 3.141592653589793  
    Dim L0 As Double  
    Dim alpha As Double  
    Dim x As Double  
  
    On Error GoTo InvalidInput  
    If PERIOD <= 0# Or DEPTH <= 0# Then GoTo InvalidInput  
  
    L0 = g * PERIOD ^ 2 / (2# * PI)  
    alpha = 4# * PI ^ 2 * DEPTH / (g * PERIOD ^ 2)  
  
    If alpha >= 20# Then  
        LWAVE = L0  
    Else  
        x = 1.199315 ^ (alpha ^ 1.047086) _  
            * alpha ^ 0.499947  
        LWAVE = L0 * Application.WorksheetFunction.Tanh(x)  
    End If  
    Exit Function  
  
InvalidInput:  
    LWAVE = CVErr(xlErrNum)  
End Function
```

For example, with $T = 9$ s and $d = 5$ m, LWAVE(9.0, 5.0) gives **L = 60.4 m**. In Excel, enter =LWAVE(A1,B1) with 9 in A1 and 5 in B1; use a semicolon instead of a comma if required by the regional settings. Save the workbook as .xlsm to retain the VBA function.